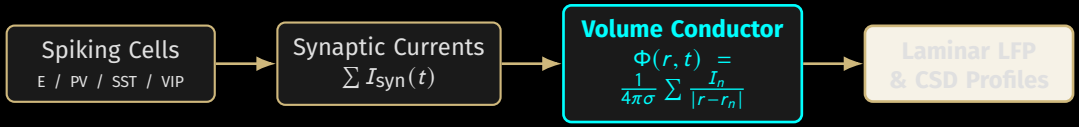
JaxFNE: Tensor-Field Neural Equations

Oscillations, Local Field Potentials, and Spectrolaminar Dynamics

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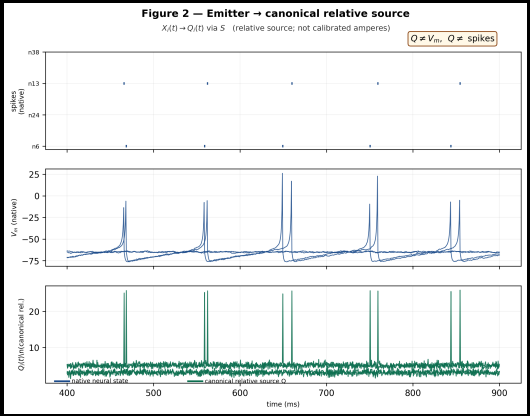
October 29, 2026

Biophysical Bridge: From Synaptic Currents to Extracellular LFP



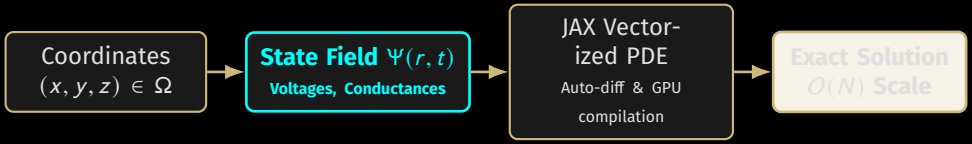
Local field potentials arise predominantly from synchronized transmembrane synaptic currents.

In JaxFNE: Neural State to Relative Source Q



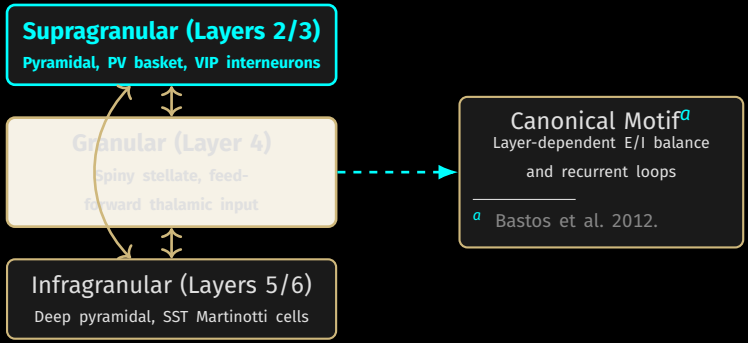
Spikes and V_m of a simulated network (top) map to the relative source $Q_i(t)$ (bottom). Source: JaxFNE manuscript v0.4.17 (Nejat), Fig. 2, experiment A. Status: claim level DEMONSTRATED (generation receipt).

JaxFNE Formulation: Continuous Field Dynamics in JAX



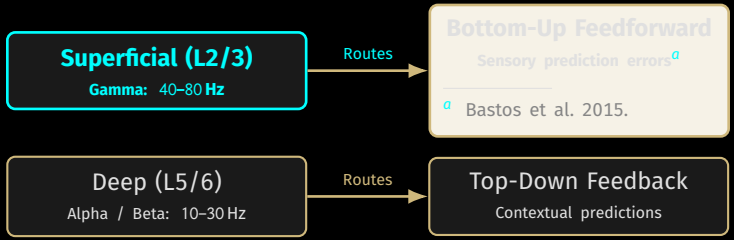
Tensor fields express continuous laminar depths without discrete connectivity bottlenecks.

Canonical Microcircuit: Layer-Specific Cortical Compartments



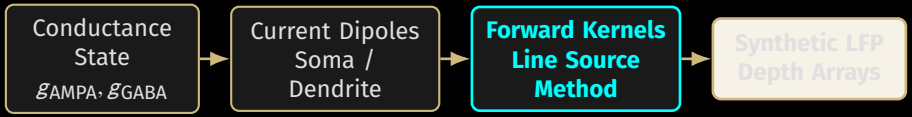
Distinct laminar compartments enforce strict division of computational labor.

Spectrolaminar Segregation: Distinct Frequency Channels



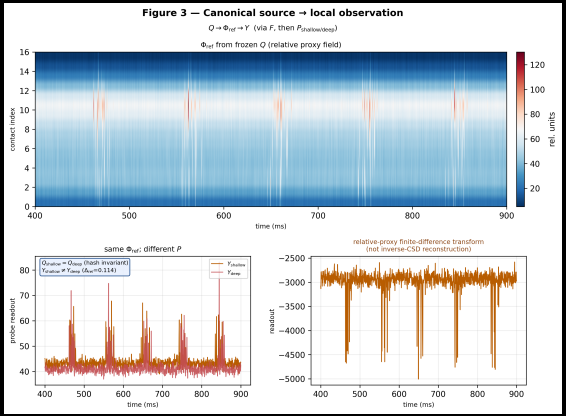
Electrophysiological power profiles mirror structural cortical hierarchy across primates.

Forward LFP Synthesis: Multi-Compartment Dipoles



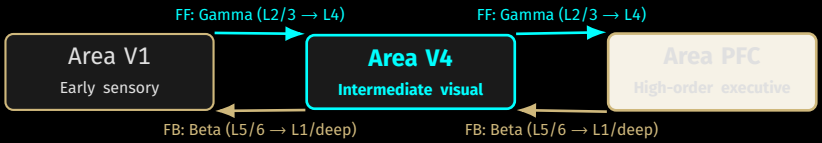
Direct resolution of extracellular field signatures from simulated neural state fields.

In JaxFNE: Source to Laminar Field and Probe Readout



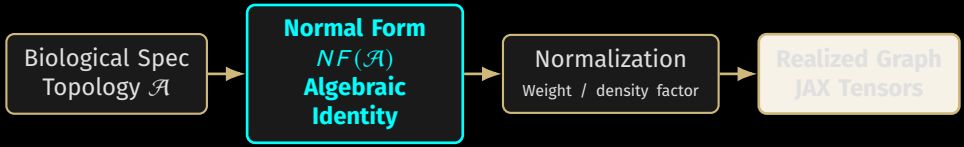
The same source Q gives a proxy field Φ_{ref} over 16 contacts (top); shallow and deep probes read it differently (bottom). Source: JaxFNE manuscript v0.4.17 (Nejat), Fig. 3, experiment A. Status: claim level DEMONSTRATED (generation receipt).

Cortical Hierarchy: Laminar Feedforward and Feedback Interplay



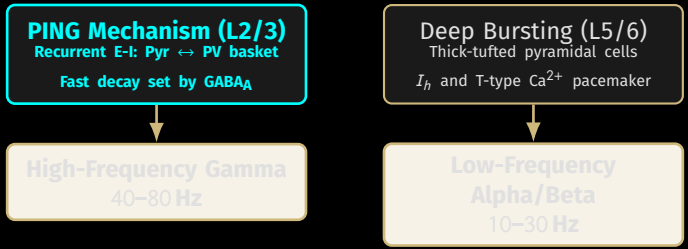
Opposing frequency streams coordinate distributed cortical processing.

TFNE Specification: Topology, Normal Form, and Compilation



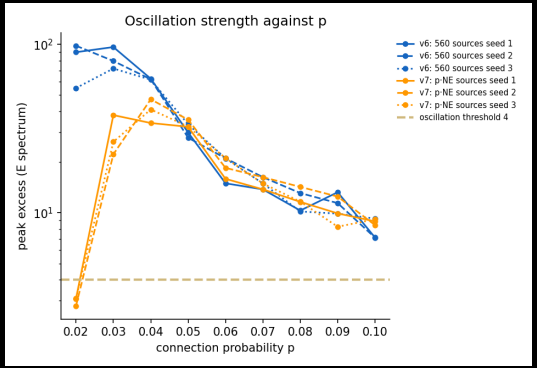
Strict separation: structural biology introduces definitions, while compilation receipts track realization.

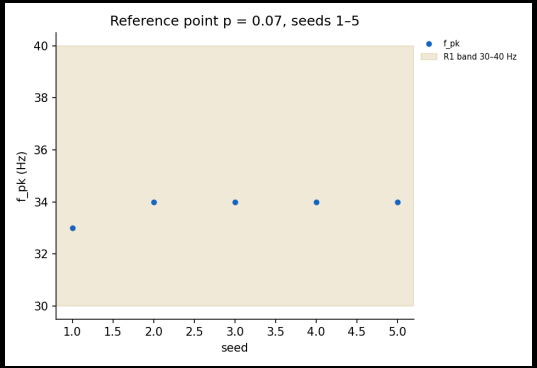
Rhythm Genesis: PING vs. Deep Intrinsic Pacemakers



Different biophysical cellular mechanisms drive superficial vs. deep frequency bands.

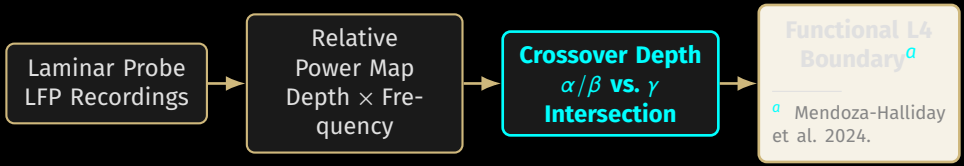
In JaxFNE: Gamma in an E–PV–SOM Network





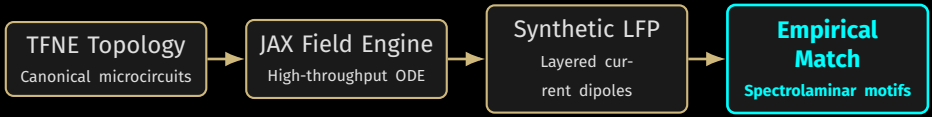
Left: oscillation strength (peak excess of the E spectrum) against connection probability p. Right: peak frequency per seed at $p = 0.07$. Source: Tahvili, Vinck and di Volo, PLoS Comput Biol 2026 (CAMINOS reference model, Fig. 1), replicated in JaxFNE. Status: mixed.

Spectrolaminar Alignment: The FLIP Coordinate Mapping



Provides an objective, electrophysiological coordinate axis for cross-area cortical comparison.

Summary: Mechanistic Synthesis in JaxFNE



Computational Strength

Continuous field equations with scalable JAX hardware acceleration.

Empirical Fidelity

Reproduces laminar oscillatory and spectrolaminar biomarkers directly.